

# Convex Decision Boundary Design for Guidance Feasibility Check During Powered Descent Phase of Chandrayaan-3 Lander

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**Abstract:** The mathematical formulation of a boundary between the guidance feasible region and guidance infeasible region of the state vector data set, aiming for zero tolerance for the infeasible region is one of the major requirements of any planetary landing guidance design problem. In this paper, a convex optimization technique is proposed to implement a mathematical formulation of the boundary between the feasible and infeasible conditions of landing, given the boundary data set of a lander. The effectiveness of the proposed method is demonstrated through an example followed by applying this method to the boundary data set obtained from the guidance and control perturbation study of the Chandrayaan-3 lunar lander. The superiority of the proposed method is established by comparing it against the results obtained through the Least Square based unconstrained optimization technique. This work is developed as a requirement for the Guidance, Navigation and Control design of Chandrayaan-3 Moon landing mission.

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**Keywords:** Decision boundary, Convex optimization, Pinpoint landing, Guidance, Trajectory design, Lunar lander, Real time decision, Autonomous system

## 1. INTRODUCTION

Chandrayaan-3 became the first Indian lunar landing mission to soft and safe land near south pole. It landed to a final latitude and longitude of 69.37356 S and 32.31975 E which was approximately 300 m away from the initial landing site. Landing is carried out autonomously where velocity reduction is achieved through *retro propulsion*. The entire mode is called Powered Descent Mode (PDM). Powered descent is initiated at perilune where the lander has orbital velocity of approximately 1.7 km/s. Through closed loop guidance, powered descent is initiated to perform landing at a pre-defined landing site. PDM is composed of Horizontal Braking phase, where the majority of lander horizontal velocity is reduced. Inertial navigation system (INS) provides lander module's position and velocity to the guidance system. The horizontal braking phase is followed by Precision Braking phase where INS is aided

by absolute sensors to provide actual position and velocity with respect to the lunar surface. Errors in INS, thrust performance deviations, mass uncertainty etc. can lead to significant deviation from nominal mission under worst cases. In case of such large perturbation, it may not be altogether feasible for to perform pin-point landing at given site. In such cases, an intelligent scheme is required to be augmented in autonomous landing sequencer that can categorize states from which pin-point landing is feasible or not. Such a decision system can then be used to switch to a contingency scheme to achieve soft landing by relaxing the pin-point landing requirement. Lander position, velocity and mass constitute the decision variables. Then feasible set is defined by a collection of all such decision variables for which closed-loop guidance converges to pre-defined target. The infeasible set is defined as a collection of all decision variables for which either guidance does not converge to terminal conditions or violate any state con-

straints. The goal is then to formulate a decision boundary for classification between feasible and infeasible set. As the onboard computer has limited memory resources, all the boundary data points cannot be stored in the memory, so the decision system need to be light-weight in terms of computation and data storage.

In Acikmese et al. (2008), *Second-order Cone Programming (SOCP)*, which is a subclass of convex programming, is suggested in the guidance algorithm for minimum fuel landing on the Martian surface with an optimized trajectory. A convex optimization-based solution that minimizes the distance to the landing site at the same time uses fuel minimally, ensuring minimum error in the case when no feasible solution exists, is proposed in Blackmore et al. (2010) for the Mars lander guidance problem. In Kitahara et al. (2017), polynomial-based guidance law for the power descending phase of lunar landing, where boundary condition is derived from approximation when the initial error occurs.

*Convex optimization* is the field of optimization where convex functions are minimized over the convex sets. Convexity makes optimization easier because local minimum is a global minimum here, hence first order conditions are sufficient to ensure optimality. In Looock et al. (2011), the *Spline approximation* problem, (given data set is approximated using smooth splines) is formulated using convex optimization. Data approximation using three methods namely coarse approximation, firefly algorithm, and convex optimization (solved using singular value decomposition) is discussed in Gálvez et al. (2013) and finally, these methods are applied to real-world problems from the CAD or CAM field. To fit multi-dimensional data, a convex piece-wise linear function with a main focus on the maximum of a fixed number of affine functions and also on its extension to general form is proposed in Magnani et al. (2009).

The paper is organized as follows. In Section 2, the requirements of convex decision boundary approach are discussed. The approach of solving the boundary design problem of a lunar lander is described in Section 3. The general structure of the optimization problem which will be used for the boundary design problem, is formulated in Section 4. The optimization problem formulated is applied to a sample problem and follows on the Chandrayaan-3 lander boundary data set in Section 5. The result obtained for the lunar lander using constrained optimization is compared with the result obtained using the Least Square technique in the same section. Section 6 presents the conclusion and future research directions.

## 2. REQUIREMENTS OF CONVEX BOUNDARY

Lander translational dynamics and associated state and control constraints can be represented as

$$\dot{r} = v \quad (1)$$

$$\dot{v} = g + a \quad (2)$$

$$\|a\| = \frac{T}{m} \quad (3)$$

$$\dot{m} = -\frac{T}{I_{sp}C} \quad (4)$$

$$T_{min} < T < T_{max} \quad (5)$$

$$\phi(t, x, \dot{x}) \leq 0 \quad (6)$$

where  $r \in R^3$ ,  $v \in R^3$  are the MCMF (Moon Centred Moon Fixed) position and velocity vectors,  $a \in R^3$  is the braking acceleration provided by propulsion thrust force  $T$  and  $m$  is the mass of lander.  $I_{sp}$  is the specific impulse of lander engines.  $g$  is the gravitational constant of the moon. For design purposes, it is taken as constant 1.62.  $c$  is the constant associated with the fuel consumption rate and is equal to 9.81.  $\phi$  represents state constraints such as, *no sub-target travel, glide-slope constraints* etc.

For a given initial and terminal conditions, there exists no analytical solution to assess if these states belong to a feasible set. It can be solved only via numerical techniques. This is not onboard-implementable due to lack of convergence guarantees, and computation limitations. An alternative approach of to this can be via data-driven methods. Through ground simulation, a data set for the lander can be constructed that comprises feasible and infeasible data points. This data set can be efficiently represented through function approximators which is then evaluated onboard based on real-time parameters. The onboard computer has a continuous log of current states such as position, velocity and mass. Based on these parameters, onboard computer needs to decide online if it is feasible to perform a safe landing at a given target.

A distinguishing criterion is essential for making the decision. Convex function approach serves as an easy way for this purpose. From the property of the convex function, evaluation of the function at a point lying inside the convex region (region bounded by the convex function) provides a value having a different sign in comparison to a point lying outside the convex region. If the point lies on the curve defined by the function, then evaluation of the function at that point results in zero. This property of convex function is utilized in this work. By storing only, the coefficients of the convex function in the onboard computer, the value of the function at current states of downrange, altitude, and mass can be obtained. From the sign of the value, it can be easily determined whether the current point lies inside the feasible region or not and a corresponding decision on landing can be taken. Hence the overall load on the onboard system gets reduced from checking the data sets against a large stored set (which is practically impossible), to a simple convex function evaluation.

## 3. PROBLEM APPROACH

The database for the landing problem for the lunar landing mission is generated from a “*Large-scale high-fidelity Monte Carlo*” simulation setup built for the actual landing mission. Simulation setup allows for starting the landing simulation from precision braking phase. Initial conditions in lander position, velocity and mass are varied on each run and simulation performance is evaluated. States that lead to successful landing are labeled as feasible and states which violate any system constraint are labeled infeasible. From the generated data set, boundary points are identified and these points are considered to design the convex boundary.

A convex function needs to be designed such that the curve defined by the function will exclude all the boundary

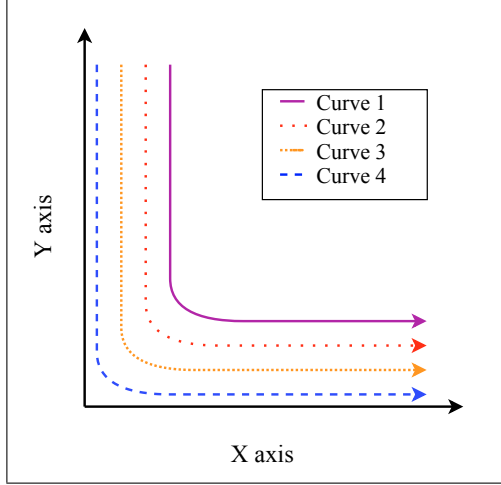


Fig. 1. Family of convex boundary curves

data points (the condition of exclusion is proposed to avoid the uncertainty in boundary data points) and lies inside the feasible region. But there exist multiple curves which satisfy the above criteria. For example, consider a problem where a boundary needs to be designed to follow the coordinate axes. In this problem, the feasible region is the first quadrant. Many convex boundaries can be fitted against the given boundary data points as shown in Fig. 1.

A criterion is required to choose to arrive at unique boundary. The selected criteria is to choose the boundary having the least L2 norm distance metric from the boundary data points so that a maximum number of data points in the feasible region can be included. This will ensure a minimum probability of false detection of infeasible states.

In Fig. 1, curve 4 has the minimum distance from the coordinate axes compare to other three curves. Therefore curve 4 serves as the best fitted boundary against the given boundary data points (coordinate axes in this case).

#### 4. PROBLEM FORMULATION

The problem of convex boundary design is formulated as a constrained optimization problem where a convex function needs to be designed such that the curve traced by the function has minimum distance from the boundary data points, excludes all of the infeasible data points and most importantly the curve remains strictly inside the feasible region. The convex function considered for this purpose has the general form of:

$$Z_0 = c_0 + c_1^T X + X^T c_2 X \quad (7)$$

where,  $c_0$  is a scalar,  $c_1 \in \mathbb{R}^n$  and  $c_2 \in \mathbb{R}^{n \times n}$  and  $n$  is the size of the feature vector  $X$ .

The value of the function (7) at a given boundary data point  $X_i$  (for any  $i$ ) is defined as the distance between the curve and  $X_i$ . By convention, coefficients are arrived such that evaluation of function at infeasible point results in negative value. Sum of all the distance values between the curve and the given boundary data points is chosen as the objective function to be minimized as shown in (8).

The curve formed by the convex function must exclude all the boundary points. Hence a minimum distance needs to be maintained between the curve and each of the data points. For ' $i$ ' th data point, this distance is denoted by  $b_i$ , where  $b_i > 0 \forall i = 1(.)N$ . Therefore for any  $X_i$ ,  $-(c_0 + c_1^T X_i + X_i^T c_2 X_i)$  should be greater than or equal to  $-b_i$  always or in other words,  $c_0 + c_1^T X_i + X_i^T c_2 X_i \leq b_i$  must be satisfied. This is formulated as inequality constraint as shown in (8).

Hence the general form of the optimization problem for convex boundary design is formulated as follows,

$$\begin{aligned} & \text{Minimize} \quad - \sum_{i=1}^N c_0 + c_1^T X_i + X_i^T c_2 X_i \\ & \text{Subject to} \quad c_0 + c_1^T X_i + X_i^T c_2 X_i \leq b_i, \forall i = 1(.)N \end{aligned} \quad (8)$$

where,  $N$  is the number of data points considered in the problem and  $c_0, c_1, c_2$  are the parameters to be designed by optimization.

#### 5. OPTIMIZATION RESULTS AND PERFORMANCE ANALYSIS

The optimization problem formulated in section 4 is in standard linear programming format and solved using *Karmarkar's algorithm*, Saxena (2017).

In this paper, the convex boundary design method is first demonstrated for a sample problem and then for the simulated data set of Chandrayaan-3 lander.

##### 5.1 Sample Problem

Consider, the intersection of a straight line and a parabola as a sample problem. The curve generated by the intersection provides the boundary data set for which the boundary needs to be fitted. The equation of the straight line is given by,

$$y = 30x - 300 \quad (9)$$

and the equation of parabola is as follows,

$$y = \frac{1}{20}x^2 \quad (10)$$

where,  $x, y$  are greater than zero. The boundary data set and the corresponding feasible region are represented in Fig. 2.

**Formulation of the Optimization Problem** The optimization problem (8) is formulated for  $n$  dimensional space. Since the data set in this problem is two dimensional that is, vector  $X$  has two features in this problem denoted by,  $X = [x, y]^T$ . Therefore the optimization problem (8) can be simplified further. Let us consider,  $c_1 = [a_1, a_2]^T$ ,  $c_2 = \begin{bmatrix} a_3 & 0 \\ a_5 & a_4 \end{bmatrix}$  and denote  $c_0$  by  $a_0$ . Substituting these values in (7), the convex function for this problem is simplified as,

$$Z_0 = a_0 + a_1 x + a_2 y + a_3 x^2 + a_4 y^2 + a_5 x y \quad (11)$$

Hence the optimization problem for the two dimensional boundary data-set is given by,

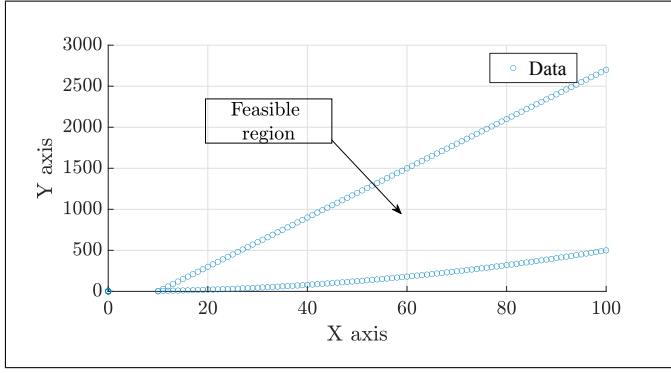


Fig. 2. Two dimensional boundary data set

$$\begin{aligned} \text{Minimize} \quad & - \sum_{i=1}^N (a_0 + a_1 x_i + a_2 y_i + a_3 x_i^2 \\ & + a_4 y_i^2 + a_5 x_i y_i) \\ \text{Subject to} \quad & AC \leq B \end{aligned} \quad (12)$$

where  $C = [a_0, a_1, a_2, a_3, a_4, a_5]^T$  contains the parameters to be tuned,  $A = \begin{bmatrix} 1 & x_1 & y_1 & x_1^2 & y_1^2 & x_1 y_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_N & y_N & x_N^2 & y_N^2 & x_N y_N \end{bmatrix}$  is a  $N \times 6$

matrix and  $B = [b_1, b_2, \dots, b_N]^T$  is a  $N \times 1$  column vector.  $N$  is the number of data points in this sample problem.

**Performance Evaluation** Matlab optimization tool `fmincon` has been used to solve the optimization problem (12). This utilizes interior point algorithm for design of  $C$ . In this problem, every element of vector  $B$  is chosen as 8. Optimized value of  $C$  is listed below:

$C = [-17.33, 1, -0.036, -0.017, -6.729 \times 10^{-5}, 0.0024]^T$ , ( $C$  is in normalized form). Designed convex function is plotted in Fig. 3 against the actual data-set generated from (9) and (10). It can be observed from Fig. 3 that, the fitted curve tries to follow the boundary points at the same time lies completely inside the feasible region, hence the objective of design is achieved.

### 5.2 Boundary Design Problem of Chandrayaan-3 Lander

The data set of the boundary region between feasible and infeasible regions of the Chandrayaan-3 lander is three-dimensional. It has the features of *downrange*, *altitude*, and *mass* respectively. So,  $X$  has three components and denote  $X$  by  $X = [x, y, z]^T$ . The data set is plotted and the corresponding feasible region is shown in Fig. 4.

**Formulation of the Optimization Problem** The optimization problem formulation is followed from 8. Let  $c_1 = [a_1, a_2, a_3]^T$ ,  $c_3 = \begin{bmatrix} a_4 & 0 & a_9 \\ a_7 & a_5 & 0 \\ 0 & a_8 & a_6 \end{bmatrix}$ . Also, denote  $c_0$  by  $a_0$ .

Using these values in (7), the convex function is reduced to,

$$\begin{aligned} Z_0 = & a_0 + a_1 x + a_2 y + a_3 z + a_4 x^2 + a_5 y^2 + a_6 z^2 \\ & + a_7 xy + a_8 yz + a_9 zx \end{aligned} \quad (13)$$

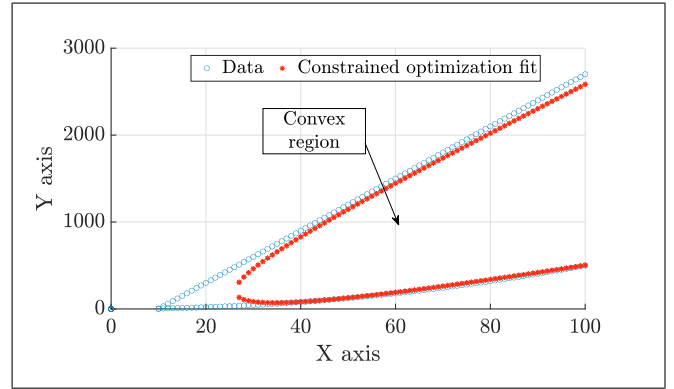


Fig. 3. Two dimensional convex boundary

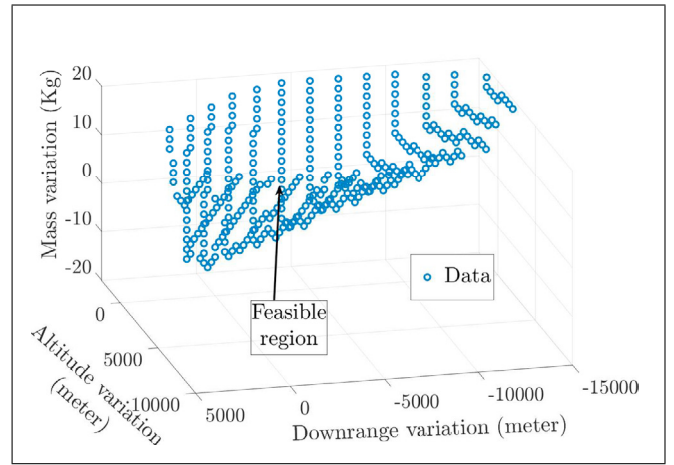


Fig. 4. Chandrayaan-3 lander boundary data set

Hence the optimization problem for lunar lander boundary data is given as follows:

$$\begin{aligned} \text{Minimize} \quad & - \sum_{i=1}^N (a_0 + a_1 x_i + a_2 y_i + a_3 z_i + a_4 x_i^2 \\ & + a_5 y_i^2 + a_6 z_i^2 + a_7 x_i y_i + a_8 y_i z_i \\ & + a_9 z_i x_i) \\ \text{Subject to} \quad & AC \leq B \end{aligned} \quad (14)$$

where,  $C = [a_0, a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9]^T$  needs to be designed via optimization,  $N$  is the number of boundary data points of the lunar lander,  $B = [b_1, b_2, \dots, b_N]^T$  and

$$A = \begin{bmatrix} 1 & x_1 & y_1 & z_1 & x_1^2 & y_1^2 & z_1^2 & x_1 y_1 & y_1 z_1 & z_1 x_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_N & y_N & z_N & x_N^2 & y_N^2 & z_N^2 & x_N y_N & y_N z_N & z_N x_N \end{bmatrix}$$

is a  $N \times 10$  matrix.

**Performance Evaluation** The optimization problem (14) is solved using the Matlab optimization tool with interior point algorithm as the option. Based on practical consideration, all the elements of vector  $B$  are selected as 4. The optimization result is as follows,  $C =$

$[0.0205, 0.0227, -0.0259, -0.382, 6.30 \times 10^{-6}, 1.439 \times 10^{-5}, 1, -9.586 \times 10^{-6}, -0.006405, 0.0049]^T$ , ( $C$  is normalized). Apart from solving the problem using optimization, the least square method (unconstrained approach) is also applied to find the best-fit curve for the given data set. Corresponding curves are shown in Fig. 5 and Fig. 6 respectively.

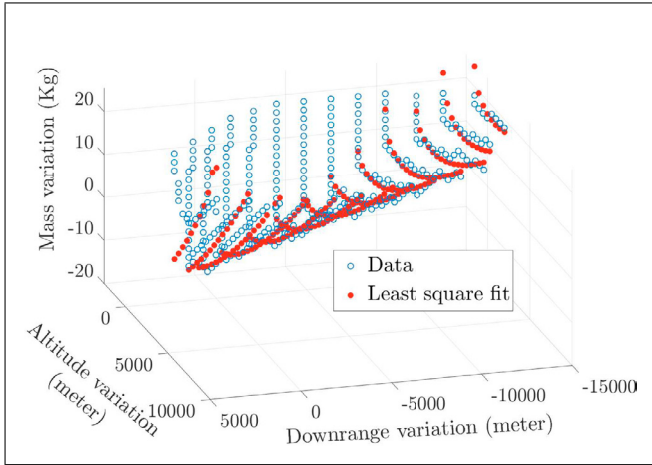


Fig. 5. Three dimensional convex boundary using Least Square technique

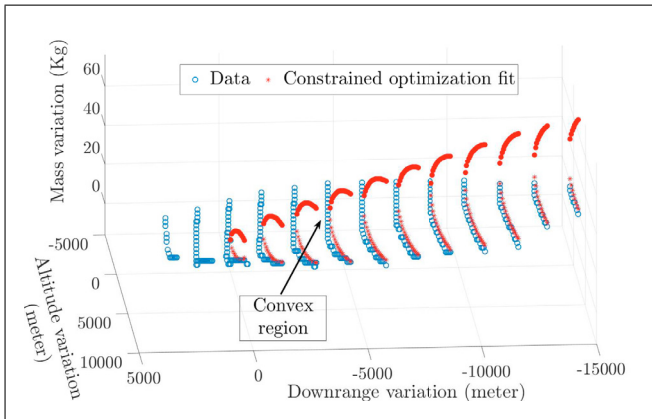


Fig. 6. Three dimensional convex boundary using Constrained Optimization technique

**Comparison of Results** Fig. 5 and Fig. 6 show that least square fit does not lie completely inside feasible region and exceeds to infeasible region whereas convex boundary designed by interior point method, remains completely inside the feasible region. This is because least square problem does not consider the minimum distance constraint from the boundary. Therefore it can be concluded that the curve formulated by constrained optimization describes the boundary data set in the best way and this method also gives the advantage of less memory usage of the onboard computer by storing only ten coefficients of the convex function (13). From the mission point of view, the designed convex function (13) is evaluated at the current data point which consists of downrange, altitude, and mass values, and sign of the result is used in decision-making. If the sign is positive, the data point is declared as feasible for nominal mission

operation, otherwise infeasible. In such case, Autonomous landing sequencer adopts a contingency plan for soft landing.

## 6. CONCLUSION

In this paper, the convex boundary design problem for the Chandrayaan-3 lunar lander data set was addressed. The curve traced by the convex function designed in this work closely followed the boundary data points of the Chandrayaan-3 lander and remained inside the feasible region at the same time. The constrained optimization method provided a good fit for the boundary points.

The future direction of the research work is as follows: (i) To apply different convex optimization algorithms for the boundary design problem of the Chandrayaan-3 lander (in this work interior point algorithm was only considered), compare the results, and choose the best performing algorithm. (ii) In the recommended solution of the boundary design problem, there are few distinct regions in the boundary data set, where no trace of convex function is present to follow the data points. In other words, there is no real solution for these regions. So, one modification that can be proposed over the existing technique is to separately apply optimization for each of these distinct regions and obtain a convex function for each region followed by the application of the least Square technique to construct a single convex function from the set of convex functions. (iii) Extend the problem to not just identify the boundary but to suitably modify the target location/guidance strategy in case system states are lying in infeasible region. (iv) Make the algorithm onboard implementable and address the problem faced in lunar missions.

## ACKNOWLEDGEMENTS

The authors of this paper would like to convey gratitude to Dr. MS Siva Group Director CDDG, Dr. Veeramuthuvel P Project Director Chandrayaan-3, U. R. Rao Satellite Centre, Bengaluru, for their support and valuable suggestions during this work. Last but not least, a special vote of thanks to the excellent peer review committee at ISRO for the Chandrayaan mission series who reviewed this work and provided valuable feedbacks regarding this research.

## REFERENCES

- Acikmese, B., Scharf, D., Blackmore, L. and Wolf, A. (2008). Enhancements on the convex programming based powered descent guidance algorithm for mars landing. *AIAA/AAS Astrodynamics Specialist Conference and Exhibit*, p. 6426.
- Blackmore, L., Acikmese, B., and Scharf, D. (2010). Minimum-landing-error powered-descent guidance for mars landing using convex optimization. *Journal of guidance, control, and dynamics*, volume 33, no. 4, pp. 1161–1171.
- Kitahara, M., Ueno, S., and Higuchi, T. (2017). Robust polynomial guidance law for power descending phase of lunar lander. *56th Annual Conference of the Society of Instrument and Control Engineers of Japan (SICE), IEEE*, pp. 470–475.

- Van Loock, W., Pipeleers, G., De Schutter, J., and Swevers, J. (2011). A convex optimization approach to curve fitting with b-splines. *IFAC Proceedings Volumes*, volume 44, no. 1, pp. 2290–2295.
- Gálvez, A., and Iglesias, A. (2013). newlbock From nonlinear optimization to convex optimization through firefly algorithm and indirect approach with applications to cad/cam. *The Scientific World Journal*, volume 2013.
- Magnani, A., and Boyd, S., P. (2009). Convex piecewise-linear fitting. *Optimization and Engineering*, volume 10, no. 1, pp. 1–17.
- Chong,E.,K., and Zak,S., H. (2013). An introduction to optimization. *John Wiley & Sons*, volume. 76.
- Saxena, S. (2017). A simple introduction to karmarkar’s algorithm for linear programming. *arXiv preprint arXiv:1712.08328*.